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Interactive system based on artificial intelligence and robotic arm to enhance arabic sign language learning in deaf children

Imane Nedjar[1,2] · **Mohammed M'hamedi**[2]

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Abstract
Tailored support is crucial for deaf and hearing-impaired children to overcome learning difficulties, particularly during primary education. The absence of listening profoundly hinders the progression of the learning journey, as it plays a pivotal role in language acquisition. Employing assistive technology is one approach to address this issue in the field of education. This paper introduces RSA, an interactive system designed for the recognition and simulation of letters in Arabic Sign Language. Our system's objective is to enrich language learning in an engaging manner. RSA utilizes artificial intelligence to identify and recognize the gestures corresponding to Arabic letters in real-time. Additionally, the system has the capability to replicate these letters through the utilization of a robotic arm. Thanks to its simplicity, the system holds promise in enhancing the acquisition of Arabic sign language skills for deaf children.

Keywords Deaf children · Learning difficulties · Arabic sign language · Artificial intelligence · Robotics

1 Introduction

The World Health Organization reports that 34 million children worldwide suffer from deafness (WHO, 2024). Hearing loss not only affects children's speech, but it can also impact their cognition, leading to social repercussions like seclusion, reduced self-esteem, and educational difficulties (WHO, 2024; Welsh et al., 2001).

✉ Imane Nedjar
imane.nedjar@univ-tlemcen.dz

Mohammed M'hamedi
mohamed.mhamedi@univ-tlemcen.dz

1 Biomedical Engineering Laboratory, Tlemcen University, Tlemcen, Tlemcen 13000, Algeria

2 First Cycle Department, Ecole Supérieure en Sciences Appliquées de Tlemcen, ESSA-Tlemcen, BP 165 RP Bel Horizon, Tlemcen 13000, Algeria

The causes and severity of deafness vary from one child to another. It can be linked to prenatal factors like genetic predisposition or intrauterine infections. Alternatively, it may occur during the perinatal period due to oxygen deprivation during birth or severe jaundice in the neonatal stage. Other contributing factors include infections, injuries to the ear or head, medications with ototoxic effects, and deficiencies in nutrition (WHO, 2024). Furthermore, syndromic causes of deafness, such as peripheral auditory impairment or central nervous system dysfunction, may lead to the coexistence of additional disabilities, including behavioral disorders, developmental delays, deafblindness, and specific learning disabilities (Antia et al., 2009; Laughton, 1989).

Deaf children, like their hearing peers, can face educational challenges that encompass two primary obstacles. The first is the impairment of learning attributed to intellectual disability, emotional distress, environmental adversities, and economic factors (Edwards, 2010). The second challenge pertains to specific learning disabilities, including auditory processing disorder, dyslexia, and dyspraxia. According to the US Interagency Committee on Learning Disabilities (ICLD, 1987), these disabilities encompass a broad spectrum of disorders marked by significant difficulties in listening comprehension, speaking fluency, reading performance, writing, reasoning, and mathematical abilities.

Unlike their hearing peers, who acquire spoken language in school, deaf children learn sign language as their primary means of communication.

Typically, sign language relies on the movements of the hands, their shapes, palm orientations, and the positioning of the arms or body (Rastgoo et al., 2021). Additionally, certain sign languages incorporate lip-patterns and facial expressions to convey nuanced word meanings (Ahdal & Nooritawati, 2012).

Sign languages vary by country, with examples including **A**merican **S**ign **L**anguage (**ASL**) in the United States, Spanish Sign Language in Spain (**L**engua de **S**ignos **E**spañola, **LSE**), and **Ar**abic **S**ign **L**anguage in the Arab world (**ArSL**). Moreover, there exists an international sign language known as **I**nternational **S**ign (**IS**), utilized during international gatherings.

As cochlear implants and middle ear implants technology have transformed the lives of numerous children, facilitating their integration into society, the use of other technologies is on the rise to further enhance the lives of deaf and hard of hearing individuals.

In recent years, significant progress has been made in the development of computer-based sign language systems employing artificial intelligence by enabling the hearing-impaired to express themselves through sign language, while also facilitating comprehension for hearing individuals. Such advanced systems are designed to recognize sign language (Wadhawan & Kumar, 2020; Wei et al., 2019; Cheok et al., 2019; Lee et al., 2021; Sharma & Singh, 2021; Sreemathy et al., 2023), interpret sign language (Dabre & Dholay, 2014; Debevc et al., 2011), translate sign language (Abiyev et al., 2020; Estrada Jiménez et al., 2016; Yin & Read, 2020; Kumar et al., 2018), and present sign language (Elliott et al., 2008; Vilhjálmsson et al., 2011; Lu & Huenerfauth, 2014; McDonald et al., 2016; Ebling & Glauert, 2016).

On the other hand, several researchers have leveraged the advantages of technology to enhance the learning of sign language. This includes the development of humanoid robot-assisted systems tailored for sign language tutoring (Kose & Yorganci, 2011;

Köse et al., 2015; Kose et al., 2014; Luccio & Gaspari, 2020; Kose et al., 2012; Uluer et al., 2015) as well as educational tools dedicated to supporting the instruction of children with hearing impairments (Imashev et al., 2024). For example, (Basiri et al., 2021) introduce the utilization of a social robot for adaptively teaching Iranian Sign Language to users. They achieve this by initially adjusting the robot's logic to align with the teaching content using a fuzzy logic mechanism during training sessions. Subsequently, the robot's teaching methods are intelligently adapted based on the user's past and present performance. Kose et al. (2014) introduced the utilization of interactive humanoid robot-based approaches for learning sign language. The primary goal of these interactive methods is to offer guidance to children with hearing impairments, aiding them in acquiring vocabulary from the Turkish sign language. The chosen set encompasses 10 signs: Spring, Throw, I/Me, Big, Mountain, Table, Black, Come, Baby, and Mother. Jemni and Elghoul (2008) have developed a web-based tool for creating multimedia courses aimed at teaching and learning sign language. This tool employs avatars to animate words extracted from the sign language dictionary. Another application, built upon educational games, was created by Kurniawan et al. (2022) to facilitate English learning for students with hearing and speech impairments.

Nawshin et al. (2020) introduced a model designed to assist children with hearing disabilities on their journey of independent reading and writing. This model employs Optical Character Recognition (OCR) to convert PDFs of paper books into sequences of Bangla sign language. Knoors and Marschark (2014) state that early hearing loss and limited communication can have a negative impact on cognitive and social-emotional development, potentially hindering learning. Nevertheless, interactive learning plays a significant role in enhancing a child's cognitive skills. Moreover, enjoyable and playful activities are known to be important for child development (Kose et al., 2014).

For this reason, in this work, we propose an interactive educational system that utilizes artificial intelligence and robotics to recognize and simulate letters of the Arabic sign language. We have named this system **RSA**, representing **R**ecognition and **S**imulation of **A**rabic sign language. The primary objective of RSA is to enhance the learning experience of Arabic sign language, making it more accessible and enjoyable through the incorporation of interactive and entertaining elements.

The article is organized as follows: Section 2 presents the Material and Method used. The evaluation results are presented in Section 3 and discussed in Section 4, Section 5 concludes the article.

2 Material and method

2.1 Proposed system

Children often have a short attention span and can be easily overwhelmed when exposed to a subject as complex as language signs. To sustain their attention and motivation for a longer period, we introduce the interactive system **RSA** for learning the letters of Arabic sign language.

RSA encompasses two key functionalities:

Gesture Recognition: this functionality leverages artificial intelligence to recognize and identify the gestures associated with Arabic letters in sign language.
Letter Simulation: the second feature involves the simulation of these letters using a robotic arm.

In RAS, there are two modules: the Recognition Module and the Simulation Module, as illustrated in Fig. 1. In the following sections, we will explain each module separately.

2.2 Recognition module

RSA is an interactive system that enables real-time recognition of letters in Arabic sign language, empowering deaf children to correlate signs with their respective letters. In essence, the children perform a sign, which is then captured by a camera. The resulting image undergoes processing by the recognition module and subsequently displays the detected letter on a display device.

To achieve the identification and recognition of letters in Arabic sign language, we used Convolutional Neural Networks (CNNs). CNNs have been widely applied in computer vision tasks and have proven to be highly effective (Bhatt et al., 2021; Nedjar et al., 2022; Schaffter et al., 2020).

We have installed the proposed system on Raspberry Pi to ensure real-time detection and classification. Raspberry Pi is a small-sized nano-computer, that functions similarly to a standard computer but is reduced to its simplest form in terms of composition.

2.2.1 Recognition model

Out of the numerous CNN architectures available, we have selected MobileNet (see Fig. 2) as basic recognition model, primarily because of its lightweight nature (Sandler et al., 2018). This specific feature makes it highly suitable for embedded systems with limited resources.

We have employed transfer learning (Iman et al., 2023) by initializing MobileNet with pre-trained ImageNet weights. In addition, we have applied fine-tuning (Peng & Wang, 2020) by adding the following layers:

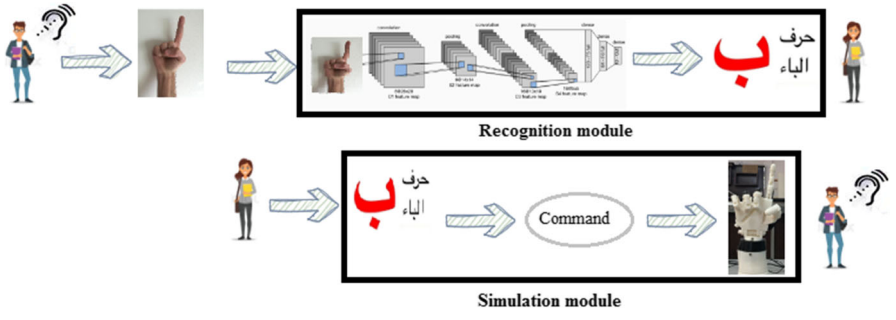


Fig. 1 Recognition and simulation modules in RSA

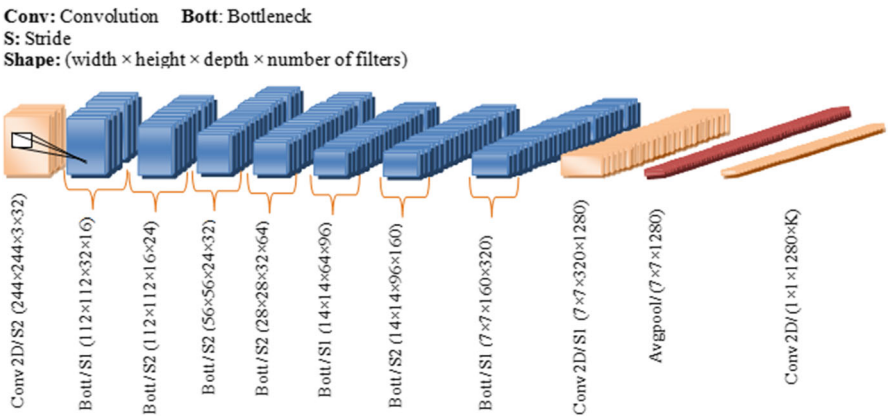


Fig. 2 MobileNet-V2 architecture

Layer= GlobalAveragePooling2D()(MobileNet.output)
Layer= Dense(1024, activation='relu')(Layer)
Layer= Dropout(0.5)(Layer)
Layer= Dense(512, activation='relu')(Layer)
Layer= Dropout(0.5)(Layer)
predictions = Dense(13, activation='softmax')(Layer)

The convolutional neural network operates through convolutions between the output of the preceding layer and the convolutional kernel (Wang et al., 2020).

$$O_j^l = \sum_{i \in M_j} X_j^{l-1} C_{ij}^l + b_l^j \tag{1}$$

$$X_j^i = f(O_j^l) \tag{2}$$

where
j represents the channel of the actual layer l
i represents the channel of the previous layer $l-1$
X_j^{l-1} denotes the output feature map of the previous layer
C_{ij}^l refers to the convolution kernel matrix
b is the bias term
f represents the activation function

2.2.2 Model configuration

Loss function we have utilized the cross-entropy loss function (F) for both training and testing stages.

$$F = -(y \log(p) + (1-y) \log(1-p)) \tag{3}$$

where
p is the predicted probability of observation.

y takes a value of 0 or 1, indicating whether a class label is the correct classification for given observation.

Optimizer we employed the Stochastic Gradient Descent (SGD) (Zinkevich et al., 2010) optimizer with a learning rate of 0.001 and a momentum of 0.9.

Epochs we trained the model for a total of 20 epochs.

2.2.3 Dataset

The recognition model was developed to classify 13 letters of ArSL, as depicted in Fig. 3. These same letters are also the ones that our robotic arm is capable of simulating.

We have utilized the RGB Arabic Alphabets Sign Language Dataset, proposed by Al-Barham et al. (2023) to train and test the recognition model. This dataset consists of fully labeled images that represent the letters of Arabic sign language. A total of 3,510 images were utilized, with 80% of them allocated for training the model, while the remaining 20% were reserved for testing. Figures 4 and 5 illustrates the distribution images for each class.

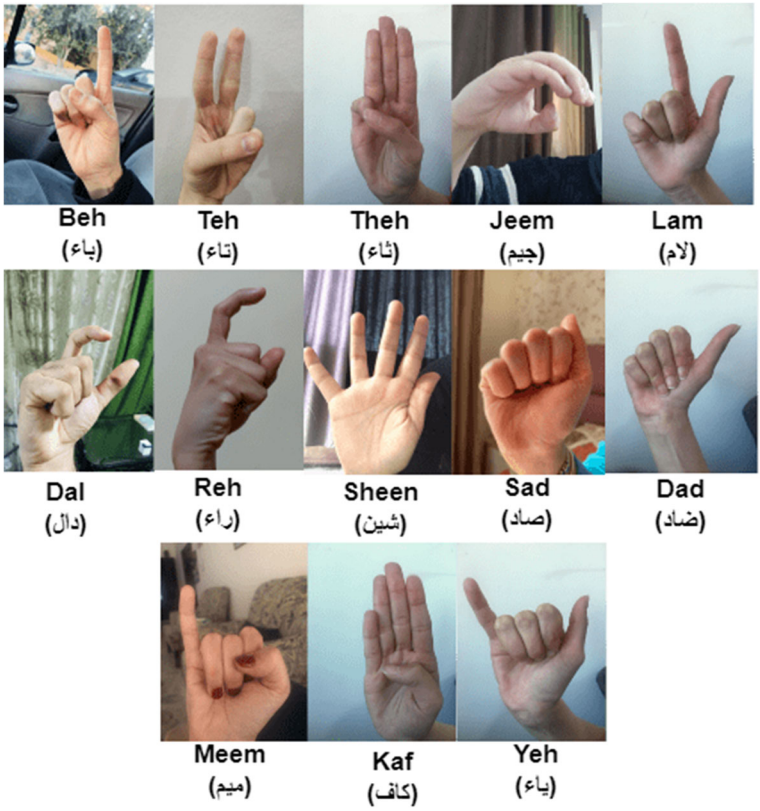


Fig. 3 The 13 letters of Arabic sign language

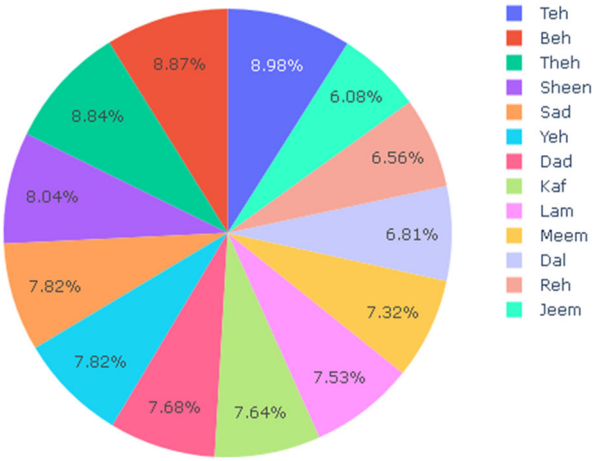


Fig. 4 The distribution of images across the 13 classes

2.2.4 Data augmentation

Data augmentation techniques are used to improve the performance of CNNs by increasing the size of the original training dataset with artificially data (Taylor & Nitschke, 2018).

In this study, we have incorporated a horizontal flip transformation, which horizontally reflects images. Furthermore, we have applied zoom transformations to magnify the images. Lastly, shearing transformations were also employed.

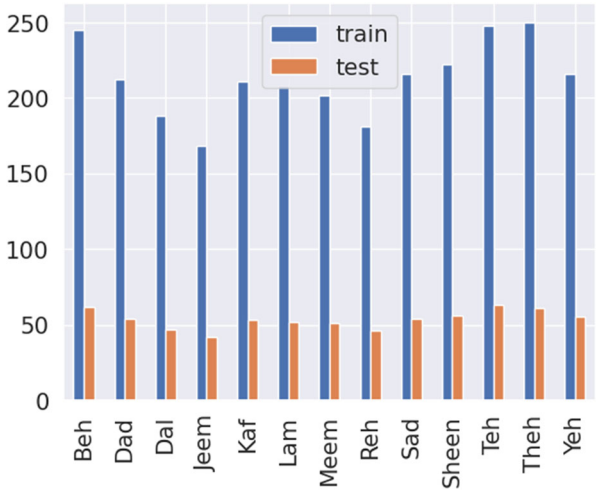


Fig. 5 The set of training and testing images for each class

2.3 Simulation module

The RSA simulation module operates by taking an Arabic language letter as input and producing the corresponding gesture in Arabic sign language. This simulation is carried out through the utilization of a robotic arm.

In the process of creating the robotic arm, we made use of the open-source 3D printed life-size robot known as InMoov (Langevin, 2024). Controlling the finger and wrist movements was achieved through the application of servomotors (specifically Metal Gears RG996R). Furthermore, for executing simultaneous movements of various fingers during the simulation, we implemented thread-based programming. In computer programming, threads serve to handle multiple concurrent actions. The electronic circuitry of RSA is illustrated in Fig. 6.

3 Results

3.1 Evaluation metrics

We evaluated the recognition model using the following metrics : accuracy, precision, recall and F1 score.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

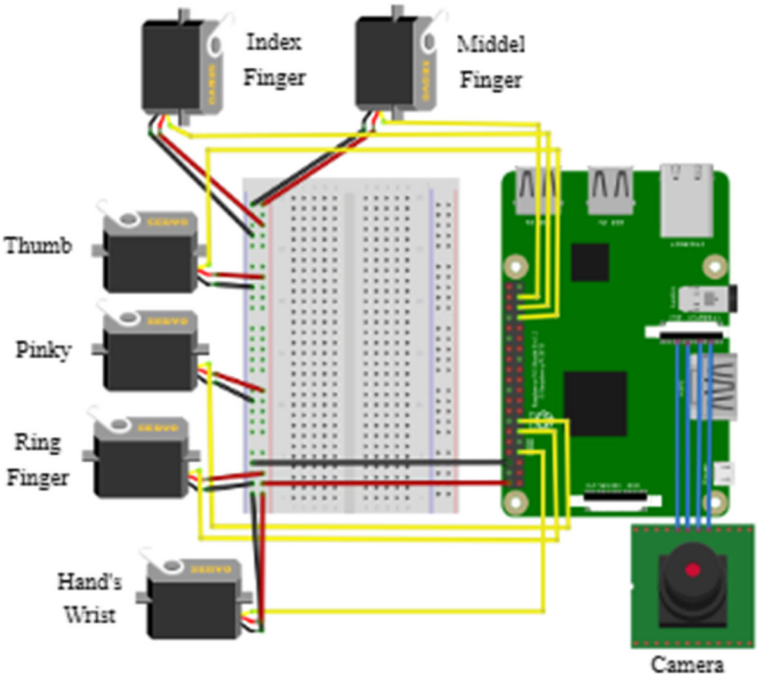


Fig. 6 The electronic circuit of RSA, created using fritzing software

$$Precision = \frac{TP}{TP + FP} \tag{5}$$

$$Recall = \frac{TP}{TP + FN} \tag{6}$$

$$F1score = 2 \times \frac{recall \times precision}{recall + precision} \tag{7}$$

where:

TP represents True Positives and TN represents True Negatives, indicating the correct classification of letters. FP represents False Positives and FN represents False Negatives, indicating the misclassification of letters.

3.2 Analysis of the results

Figure 7 illustrates the learning process throughout epochs with the presentation of loss and accuracy curves. It is evident that precision steadily improves as the epochs progress. After 20 epochs, the model achieved an impressive training accuracy of 99.28%, with a corresponding loss value of 0.0508. On the test dataset, the model exhibits remarkable performance, achieving an accuracy of 98.56%, precision of 98.54%, recall of 98.46%, and an F1 score of 98.47%.

The utilization of the confusion matrix depicted in Fig. 8 offers a comprehensive perspective on the performance of our sign language recognition model. It effectively outlines the classification outcomes for the various letters in Arabic sign language. Upon analyzing the values within the confusion matrix, it becomes apparent that the recall ranges from 89% to 100%. Notably, the letters 'Beh', 'Dad', 'Jeem', 'Lam', 'Sad', 'Sheen', 'Theh', and 'Yeh' all achieved a perfect recall value of 100%. However, the letter 'Reh' exhibited the lowest recall at 89%, with 2% of its images misclassified as 'Yeh' and 9% as 'Dal'. This can be attributed to the significant resemblance between the letter 'Reh' and the letter 'Dal' (see Fig. 3), which likely contributed to the misclassifications.

Table 1 displays the Precision, Recall, and F1-score values for each letter, calculated from the confusion matrix. Notably, the results reveal that the letters 'Beh', 'Dad', 'Jeem', 'Lam', and 'Sheen' achieved an outstanding classification accuracy of 100%.

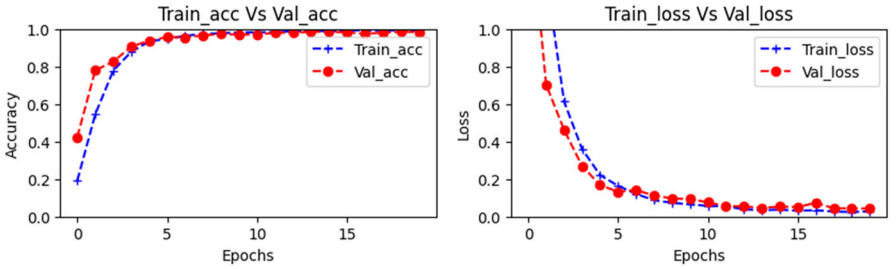


Fig. 7 The loss and accuracy curves of the model on the training and validation sets

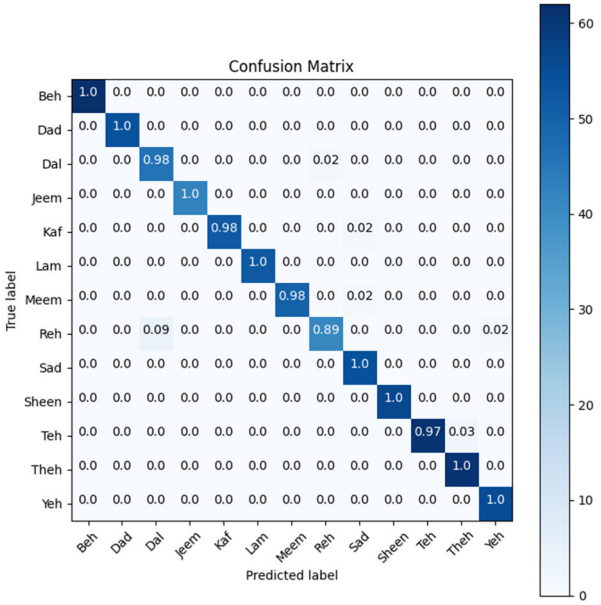


Fig. 8 Confusion matrix for the classification of the 13 letters of Arabic sign language

Furthermore, the precision for the letters 'Kaf', 'Meem', and 'Teh' reached 100%, while their respective recall values were 98%, 98%, and 97%. Moreover, the F1 scores are within the range of 93% to 100%.

Figure 9 depicts the simulation of the 13 letters of Arabic sign language using the robotic arm.

Table 1 Precision, Recall and F1-score for each class

Classes	Precision	Recall	F1-score
Beh	1.00	1.00	1.00
Dad	1.00	1.00	1.00
Dal	0.92	0.98	0.95
Jeem	1.00	1.00	1.00
Kaf	1.00	0.98	0.99
Lam	1.00	1.00	1.00
Meem	1.00	0.98	0.99
Reh	0.98	0.89	0.93
Sad	0.96	1.00	0.98
Sheen	1.00	1.00	1.00
Teh	1.00	0.97	0.98
Theh	0.97	1.00	0.98
Yeh	0.98	1.00	0.99

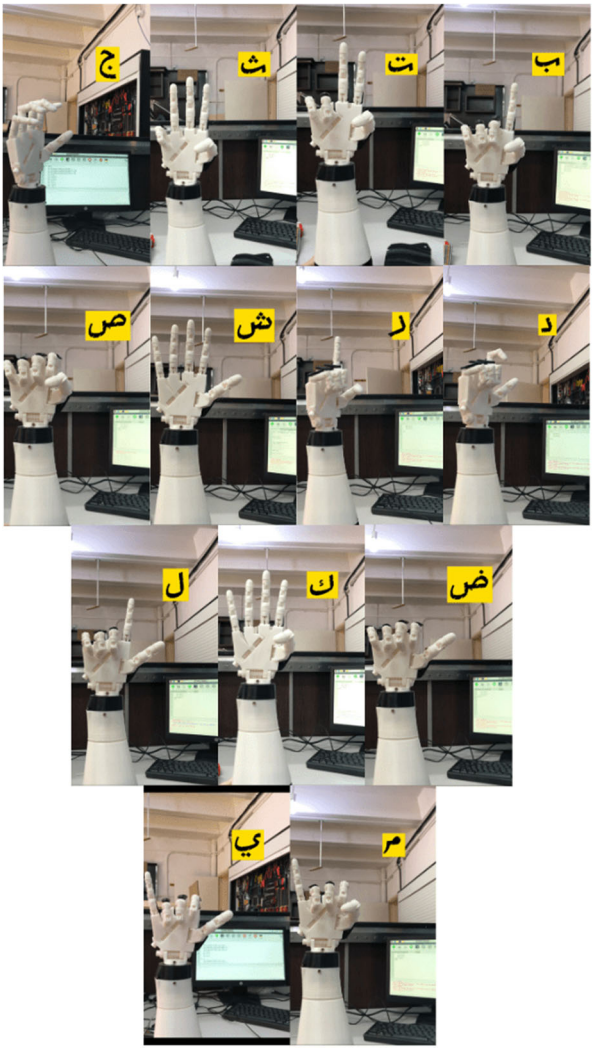


Fig. 9 The simulation of the 13 letters of Arabic sign language using the robotic arm

4 Discussion

Deaf and hearing-impaired children often face diverse learning challenges, which can result in lower academic achievement compared to their hearing peers (Marschark & Knoors, 2012). Providing early educational support for deaf children is crucial to address these obstacles and ensure effective guidance for academic success.

In this study, we introduce an interactive RSA system aimed at aiding deaf children in learning Arabic sign language letters during primary education. RSA is intended to assist children in overcoming the prevalent challenges associated with learning disabilities. Indeed, learning disabilities have frequently been linked to memory issues, visual

perception difficulties, and executive functioning challenges (Elliott et al., 1988). These can lead to specific learning disabilities such as language deficits, like dyslexia, which involve difficulties in reading and writing (Edwards, 2010).

Short-term memory (working memory) and long-term memory (semantic memory) both play crucial roles in language learning (Afraimovich et al., 2015). According to Marschark and Knoors (2012), deaf children encounter challenges in solving sequential tasks that require remembering information in a specific order. This difficulty is often attributed to what is known as sequential memory. It is suggested that this arises because deaf children fail to link information stored in short-term memory with the knowledge stored in long-term memory. In our interactive RSA system, deaf children are able to repeat the process of recognizing Arabic letters, which are subsequently simulated by a robotic arm. This serves to engage their long-term memory in memorizing these letters, while also providing practice in sequential tasks, given that the simulation follows the recognition phase.

On the other hand, deaf individuals are at a higher risk of experiencing vision impairment compared to their hearing counterparts. This is attributed to the fact that the causes of deafness can also impact the eyes (Wiley, 2012). Additionally, deaf children have a heightened sensitivity to peripheral vision, as their brains develop to allocate more attention to the visual environment. While this may seem advantageous, according to Dye et al. (2008), it can have a negative effect on the concentration of deaf children in a noisy classroom environment. To address these issues, our system incorporates a robotic arm for letter simulation, enabling deaf children to use both the tactile and visual senses.

Language is linked to individual cognition, with exclusive functions like thoughts and behaviors representing one aspect of cognitive processing. An important consideration in the learning process of deaf children is their struggle to determine whether they have acquired information, sometimes leading to overestimation of their understanding (Borgna et al., 2011; Marschark & Knoors, 2012). So, it is beneficial for them to receive adequate assistance and be provided with the best practices for self-assessment. In RSA, as the deaf children enter the letter to be simulated by the robotic arm, they can verify if the result matches their expectations, thereby reinforcing self-evaluation.

The role of parents and the environment is crucial in the learning of sign language. However, communication between deaf children and hearing parents may not always be as effective as with deaf parents, and hearing parents often struggle to teach sign language to their children, as highlighted by Razalli et al. (2019). Our primary objective in developing this system is to facilitate the learning of sign language for deaf children during their primary education, making it accessible not only in schools but also to parents. Considering that 80% of people with hearing disabilities reside in low and middle-income countries (WHO, 2024), we have taken into account the cost-effectiveness of this system, which is significantly lower than utilizing a complete robot.

In the context of Arabic Sign Language, there have been efforts directed towards the creation and organization of Arabic sign language dictionaries, along with the incorporation of technology (Al-Fityani & Padden, 2008; Balaha et al., 2023). Furthermore, a substantial portion of research and applications has been centered around ArSL recognition (Ibrahim et al., 2018; Tharwat et al., 2015; Alzohairi et al., 2018;

Kamruzzaman, 2020; Elatawy et al., 2020) and ArSL translation (Halawani, 2008; Luqman & Mahmoud, 2019; El-Bendary et al., 2010). However, there are only a few works related to enhancing Arabic sign-language learning for children with hearing impairments. As an example, Mohammad et al. (2022) introduced a mobile application specifically designed to teach Arabic sign language, catering particularly to children under 12 years old. Another mobile application developed by Al-Nafjan et al. (2015), named Tawasol, aims to facilitate the acquisition of Arabic sign language. This application includes features such as a sign dictionary and fingerspelling. Compared to the aforementioned approaches, the proposed interactive RSA system integrates artificial intelligence and robotics to enhance the learning of Arabic sign language. In fact, the system can recognize ArSL letters in real time. Furthermore, it can simulate the signs using a robotic arm. This process is more engaging, as children can physically interact with the robotic arm, treating it as a toy.

4.1 Current limitations and future prospects

The RSA system, as it stands, is tailored to offer early support for deaf children in acquiring ArSL. However, there are certain constraints that could be refined in future endeavors, such as incorporating phonological knowledge. Phonological skills are pivotal for both hearing and deaf children in fostering proficient reading abilities (Trezek & Malmgren, 2005). Furthermore, we aspire to augment the system's functionalities to encompass a wider spectrum of letters, words, and phrases. This enhancement is especially crucial given the typically less extensive vocabularies and conceptual knowledge observed in deaf children compared to their hearing peers (Marschark & Knoors, 2012).

5 Conclusion

Language acquisition poses unique challenges for children, especially those who are deaf or hard of hearing. In this study, we introduce an interactive system RSA designed to recognize and simulate letters in Arabic sign language. The aim of this system is to assist deaf children in acquiring sign language during their primary education. Specifically, RSA addresses prevalent issues in learning disabilities, including memory recall, visual perception difficulties, and executive functioning.

The system comprises two modules: one for recognition and another for simulation.

In the recognition module, we employed convolutional neural networks, specifically MobileNet, to detect the signs associated with Arabic letters. This enables deaf children to correlate the signs with the letters. We have evaluated the performance of this module using the RGB Arabic Alphabets Sign Language Dataset, achieving an accuracy of 98.56%, precision of 98.54%, recall of 98.46%, and an F1 score of 98.47%.

In the simulation module, there is a robotic arm capable of replicating the letters of Arabic sign language. This arm's movements are controlled by servo motors (Metal Gears RG996R), and we implemented thread-based programming to achieve simultaneous movement of different fingers. The RSA can be adapted to learn other sign languages by adjusting the two modules.

With robust technological support and moral encouragement, deaf children can excel in their academics and personal lives.

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Declarations

Competing Interests The authors declare no conflict of interest.

Ethical Statement Not applicable.

Informed Consent Not applicable.

Statement Regarding Research Involving Human Participants and/or Animals Not applicable.

Consent to Participate Not applicable.

Consent to Publish All coauthors have consented for publication.

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